

Course "Mathematical Programming Modelling"
(42112)
Some simple training assignments
DTU 2025

Assignment Collection

In this document we will include a few easy assignments, for preparation for the Mathematical Programming Course (42112)

Incredible Chairs

The company Incredible Chairs produces two different types of chairs, A and B , where one Chair A can be sold for a profit of 4 and one Chair B can be sold for a profit of 6. The chairs are produced on three production lines:

1. Line 1 requires 2 hour to produce one unit of Chair A , and at most 14 hours can be used due to other production on the line.
2. Line 2 requires 3 hours to produce one unit of Chair B , and at most 15 hours can be used.
3. Line 3 requires 4 hours to produce one unit of Chair A and 3 hours to produce one unit of chair 2, and at most 36 hours can be used.

The objective is to maximize the total profit, given the above mentioned profit limitations. This leads to the following LP:

$$\begin{array}{ll}\text{Maximize.} & 4 \cdot x_A + 6 \cdot x_B \\ \text{Subject to :} & \\ & 2 \cdot x_A \leq 14 \\ & 3 \cdot x_B \leq 15 \\ & 4 \cdot x_A + 3 \cdot x_B \leq 36 \\ & x_A \geq 0 \\ & x_B \geq 0\end{array}$$

On the next page, the corresponding model is given in Julia.

Compact Julia/JuMP program/model:

```
using JuMP
using HiGHS
IC = Model(HiGHS.Optimizer)

@variable(IC, xA >= 0)
@variable(IC, xB >= 0)
@objective(IC, Max, 4*xA + 6*xB)
@constraint(IC, 2*xA <= 14)
@constraint(IC, 3*xB <= 15)
@constraint(IC, 4*xA + 3*xB <= 36)
print(IC)
optimize!(IC)
println("Termination status: $(termination_status(IC))")
if termination_status(IC) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(IC))")
    println("xA: ", value(xA))
    println("xB: ", value(xB))
else
    println("No optimal solution available")
end
```

Assignment 0.1

Copy-paste the above compact IC model from the pdf file. You will probably have to clean up the code. Run the code and check the results.

Optimal objective value for Incredible Chairs Problem: 51.0 with $x_A = 5.25$ and $x_B = 5.0$

The standard format for the models in this book is a bit more expansive and is given below.

```
*****
# Incredible Chairs, Simple LP
using JuMP
using HiGHS
*****

*****
```

```

# PARAMETERS
Chairs = ["A", "B"]
C=length(Chairs)
ProductionLines = ["1","2","3"]
P=length(ProductionLines)

Profit = [4,6]
Capacity = [14,15,36]
RecourseUsage =
    [
        2 0;
        0 3;
        4 3
    ]

#####

#####
# Model
IC = Model(HiGHS.Optimizer)

@variable(IC,x[1:C]>=0)

@objective(IC, Max, sum( Profit[c]*x[c] for c=1:C))

@constraint(IC, [p=1:P], sum( RecourseUsage[p,c]*x[c] for c=1:C) <= Capacity[p])
#####

#####
# Solve
solution = optimize!(IC)
println("Termination status: $(termination_status(IC))")
#####

#####
# Solution
if termination_status(IC) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(IC))")
    for c=1:C
        println("x[" , Chairs[c] , " ] : ",value(x[c]))
    end
end

```

```

else
    println("No optimal solution available")
end
#####

```

Assignment 0.2

Copy-paste the above compact IC model from the pdf file. You will probably have to clean up the code. Run the code and check the results.

The model is **identical** with the previous one, so the optimal objective value for Incredible Chairs Problem: 51.0 with $x_A = 5.25$ and $x_B = 5.0$

NOTICE: There is no problem in copying the IncredibleChairs.jl standard program for the next assignments, BUT be care full. Remember to update all the data in the program. In particular make sure to name the sets, cardinality values and index names correctly. Here the sets are Chairs and ProductionLines. The cardinality values C and T. And the indexes are c and t.

Incredible Chairs 2

Incredible Chairs has decided to extend their line of products to 8 other chairs and now the model should include all the 5 production lines. Below we give the data: The profit for each chair is given in the table below.

	Chair types									
	A	B	C	D	E	F	G	H	I	J
Profit (€)	6	5	9	5	6	3	4	7	4	3

Table 1: Profit for each chair sold (€)

The capacity for each of the production lines is given in the table below.

	Production lines				
	1	2	3	4	5
Production line capacity	47	19	36	13	46

Table 2: Production line capacity

The resources needed for each chair on each line is given in the table below.

	Chair types									
	A	B	C	D	E	F	G	H	I	J
1	6	4	2	3	1	10	2	9	3	5
2	5	6	1	1	7	2	9	1	8	6
3	8	10	7	2	9	6	9	6	5	6
4	8	4	8	10	5	4	1	5	3	5
5	1	4	7	2	4	1	2	3	10	1

Table 3: Resource usage for each chair on each production line

Assignment 0.3

Formulate an LP to maximize the profit of the chair production

Optimal objective value for Incredible Chairs 2: 23.04 €.

Class Jobs

Your little sister and four of her friends from primary school have volunteered to do a number of different community jobs (like picking up trash etc.). The five children $c \in C$ have five jobs $j \in J$ to choose between. To make the work more attractive, each child is asked to rank their preference for each job on a scale from 1 to 5 (with five being best). This information is given in Table 4, i.e. $Wish_{j,c} \in \{1, 2, 3, 4, 5\}$. You can read about how to define data in Julia in Chapter ??.

	c_1	c_2	c_3	c_4	c_5
j_1	1	3	2	5	5
j_2	5	2	1	1	2
j_3	1	5	1	1	1
j_4	4	5	4	4	4
j_5	3	5	3	5	3

Table 4: Job preferences

Assignment 0.4

Formulate an LP that can be used to find the best assignment of the jobs to the children. The best assignment is the one that maximizes the total preference score. Each child can only do one job, and each job can only be done by one child.

Optimal solution for Class Jobs: 24

Is the solution **integer**? In other words, are all decision variable values equal to either 1 for the job assignments made, and 0 for the job assignments not made? They should be. This is a feature of this type of problem. They are generally referred to as **assignment problems**. If we add more constraints, this property is usually not guaranteed.